SIMATS ENGINEERING
ASSESSMENT TOOL 3

CO-4-AT-3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:50**

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items *(2)*	Comparison Depth(2.5)	Use of Evidence *(2)*	Critical Thinking *(2)*	Clarity of Presentation *(1)*	*Total(10)*
Weightage	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I S. chandra Shekher (192372163) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. chandra shekhar

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer	Income (₹)	Spending Score
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:
Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:
Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25

Point	**Latitude**	**Longitude**
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:
Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

Product	**Sales (₹)**	**Frequency**
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:
Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

Q5 . Dataset: Mixed Attributes

Item	**Price (₹)**	**Category**	**Rating**
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:
Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

- AT-3

Dataset : Customer Data

Customer	income	Sapding score
C1	30000	60
C2	60000	40
C3	50000	70
C4	88000	65
C5	65000	35

K-Means clustering group customer into a fixed Number of cluster based on Similarity. for dataset. it can quickly seprate customers with low income high spending and high income - low spending

Hierachical clustering

Creates a tree-like structure (dendrogram) showing how customers are ground at different level

Comparsion:-

K-means is generally faster and more scalable for large datasets while hieracical clustering is more suitable for small dataset and understanding cluster relationship.

(2) Dataset : Document Data

Document	Word1	word2	word 3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

K-Means assign each document to only one cluster (hard clustering).
For example D1, D2, and D5 may belong to one cluster while D3 and D4 form another cluster

EM (Expectation - maximization)

use soft clustering, assiging probobilites of belong to multiple cluster
∴ it handles overlapping cluster better than K-means

Conclusion

For this dataset Em, is more suitable when document cluster overlap while K-mean is simpler and faster but less effective overlapping cluster

③ Dataset : Geographic Data

point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

For the given geographic point, Euclidean distance measure the stright line distance b/w two points

$$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

Mahnhattan distance

It measure the total horizontal and vertical distance:

$$d = |x_2 - x_1| + |y_2 - y_1|$$

impact on clustering

Euclidean distance is suitable for measuring direct geographic proximity, while manhattan distance may give different cluster boundaries because it sum coordinates difference. Thus the selected distance Metric can effect which point grouped together.

(4) Dataset :- Retail products

Product	sales	frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Agglomerative clustering

It is a bottom-up approach. it start with each product as a seperate cluster and gradually merge similar product for this dataset P1 and P3 may form a high-sales cluster while P2, P4 and P5 form a lower sales cluster

Divisible clustering

It is a top-down approach. It start with all product in one cluster and repeatly divides them into smaller groups

Scuitability

For this small retail dataset, both approaches are suitable but agglomerative clustering is simple and more commonly used it can effectively group product based on similar sales and purchase fre-quency

(5) **Dataset:- Mixed Attributes**

Item	price	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Centriod based clustering (such as K-means) works manily with Numerical data and form cluster around a central point. for this dataset, it can use price and Rating but categorical (A/B) must first be converted into Numerical form

Density-based clustering (such as DBSCAN) group item based on density of nearby data point and can identify irregular cluster and Noise.

Conclusion

For this small mixed-attribute dataset, centriod-based clustering is effective after proper encoding and Normalization while density clustering is more flexible for detecting Natural groups However. mixed data type require suitable distance measured or preprocessing in both approaches

Cluster mode

- (•) Use training set
- ( ) Supplied test set Set...
- ( ) Percentage split % 66
- ( ) Classes to clusters evaluation
  - (Num) SpendingScore
- [x] Store clusters for visualization

Ignore attributes

Start | Stop

Result list (right-click for options)

- 06:44:41 - SimpleKMeans
- 06:46:07 - SimpleKMeans

Clusterer output

```
=== Clustering model (full training set) ===


kMeans
======

Number of iterations: 2
Within cluster sum of squared errors: 0.27636737675347706

Initial starting points (random):

Cluster 0: 0,65
Cluster 1: 0.864865,40

Missing values globally replaced with mean/mode

Final cluster centroids:
                             Cluster#
Attribute        Full Data          0          1
                     (5.0)      (3.0)      (2.0)
================================================
Income              0.5027     0.2162     0.9324
SpendingScore           54         65       37.5




Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      3 ( 60%)
1      2 ( 40%)
```

Cluster mode

- (•) Use training set
- ( ) Supplied test set [Set...]
- ( ) Percentage split % 66
- ( ) Classes to clusters evaluation

(Num) SpendingScore

[x] Store clusters for visualization

[Ignore attributes]

[Start] [Stop]

Result list (right-click for options)

- 06:44:41 - SimpleKMeans
- 06:46:07 - SimpleKMeans
- 06:47:37 - HierarchicalClusterer

Clusterer output

```
=== Run information ===

Scheme:       weka.clusterers.HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Relation:     customers-weka.filters.unsupervised.attribute.Normalize-S1.0-T0.0-weka.filters.unsupervised.attribute.Remove-R1
Instances:    5
Attributes:   2
              Income
              SpendingScore
Test mode:    evaluate on training data


=== Clustering model (full training set) ===

Cluster 0
((60.0:0.15274,65.0:0.15274):0.45866,70.0:0.61141)

Cluster 1
(40.0:0.19665,35.0:0.19665)



Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      3 ( 60%)
1      2 ( 40%)
```

Cluster mode

- (•) Use training set
- ( ) Supplied test set — Set...
- ( ) Percentage split — % 66
- ( ) Classes to clusters evaluation — (Str) Document
- [x] Store clusters for visualization

Ignore attributes

Start | Stop

Result list (right-click for options)

- 06:51:49 - SimpleKMeans
- 06:51:58 - SimpleKMeans
- 06:52:07 - SimpleKMeans

Clusterer output

```
kMeans
======

Number of iterations: 2
Within cluster sum of squared errors: 0.17523053665910812

Initial starting points (random):

Cluster 0: 0.928571,0.1,0.1
Cluster 1: 0,1,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:
                          Cluster#
Attribute     Full Data          0          1
                  (5.0)      (2.0)      (3.0)
===============================================
Word1            0.4429     0.9643     0.0952
Word2               0.5       0.05        0.8
Word3              0.24        0.1     0.3333




Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0       2 ( 40%)
1       3 ( 60%)


Class attribute: Document
```

Cluster mode

- (●) Use training set
- ( ) Supplied test set [illegible]
- ( ) Percentage split % [illegible]
- ( ) Classes to clusters evaluation

(Nom) Document

[✓] Store clusters for visualization

Ignore attributes

Start | Stop

Result list (right-click for options)

- [illegible] - SimpleKMeans
- [illegible] - SimpleKMeans
- [illegible] - SimpleKMeans
- [illegible] - HierarchicalClusterer

Clusterer output

```
            Document
Test mode:  Classes to clusters evaluation on training data

=== Clustering model (full training set) ===

Cluster 0
[illegible]

Cluster 1
[illegible]


Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0       [illegible]
1       [illegible]


Class attribute: Document
Classes to Clusters:

 0 1  <-- assigned to cluster
 [illegible]

Cluster 0 <-- [illegible]
Cluster 1 <-- [illegible]

Incorrectly clustered instances :   [illegible]
```

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Clusterer

Choose | **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1

Cluster mode

- Use training set
- Supplied test set — Set...
- Percentage split — % 66
- Classes to clusters evaluation
  - (Num) Longitude
- Store clusters for visualization

Ignore attributes

Start | Stop

Result list (right-click for options)

06:58:54 - SimpleKMeans

Clusterer output

```
======

Number of iterations: 4
Within cluster sum of squared errors: 0.2482445759368771

Initial starting points (random):

Cluster 0: 0.615385,80.3
Cluster 1: 1,80.25

Missing values globally replaced with mean/mode

Final cluster centroids:
                          Cluster#
Attribute     Full Data          0          1
                  (5.0)      (2.0)      (3.0)
=============================================
Latitude         0.5385     0.1154     0.8205
Longitude        80.248      80.21    80.2733




Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0       2 ( 40%)
1       3 ( 60%)
```

Log

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.ManhattanDistance -R first-last" -I 500 -num-slots

Cluster mode

- (•) Use training set
- ( ) Supplied test set [Set...]
- ( ) Percentage split % 66
- ( ) Classes to clusters evaluation
  (Num) Longitude
- [✓] Store clusters for visualization

Ignore attributes

Start | Stop

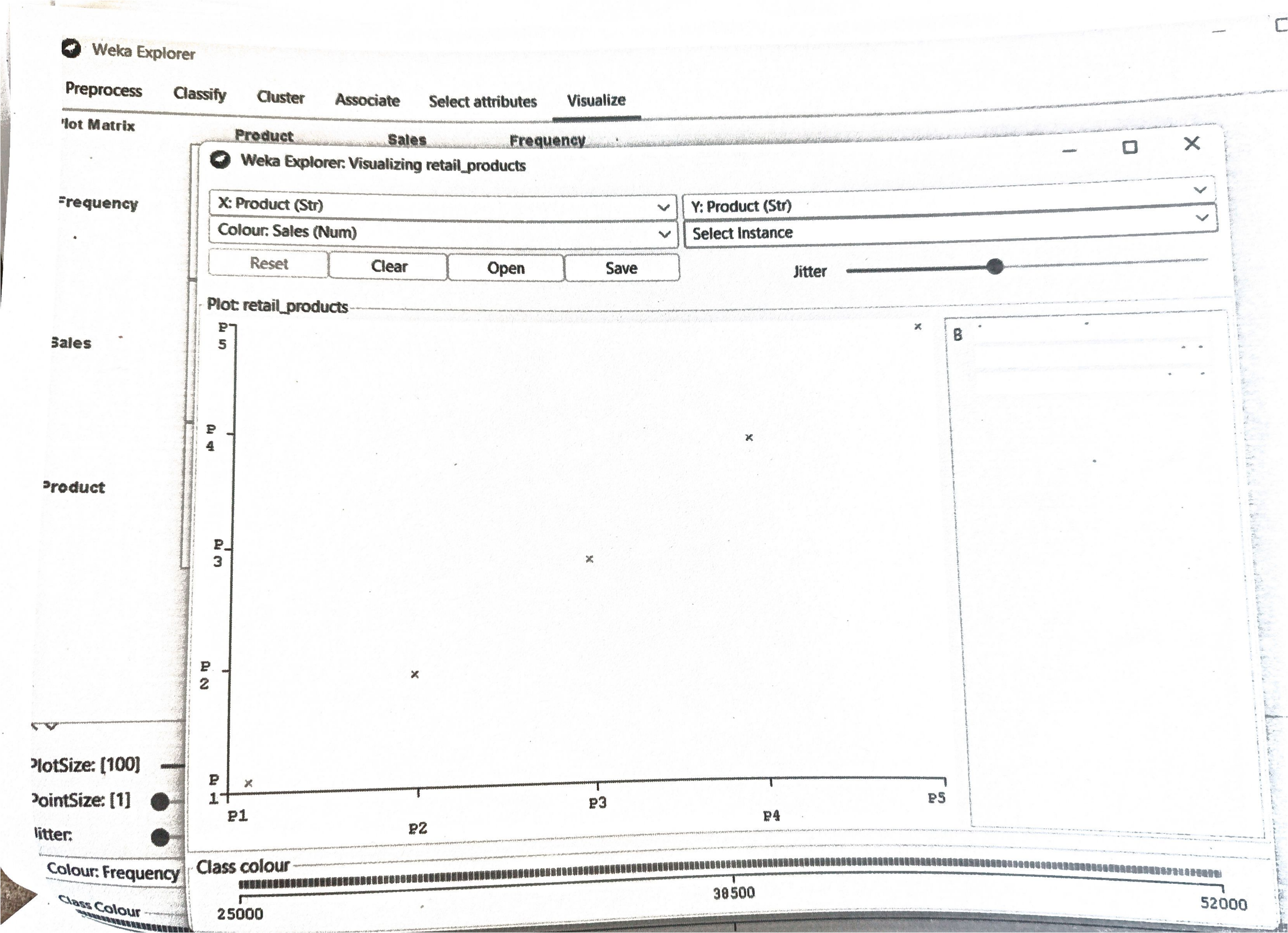
Result list (right-click for options)

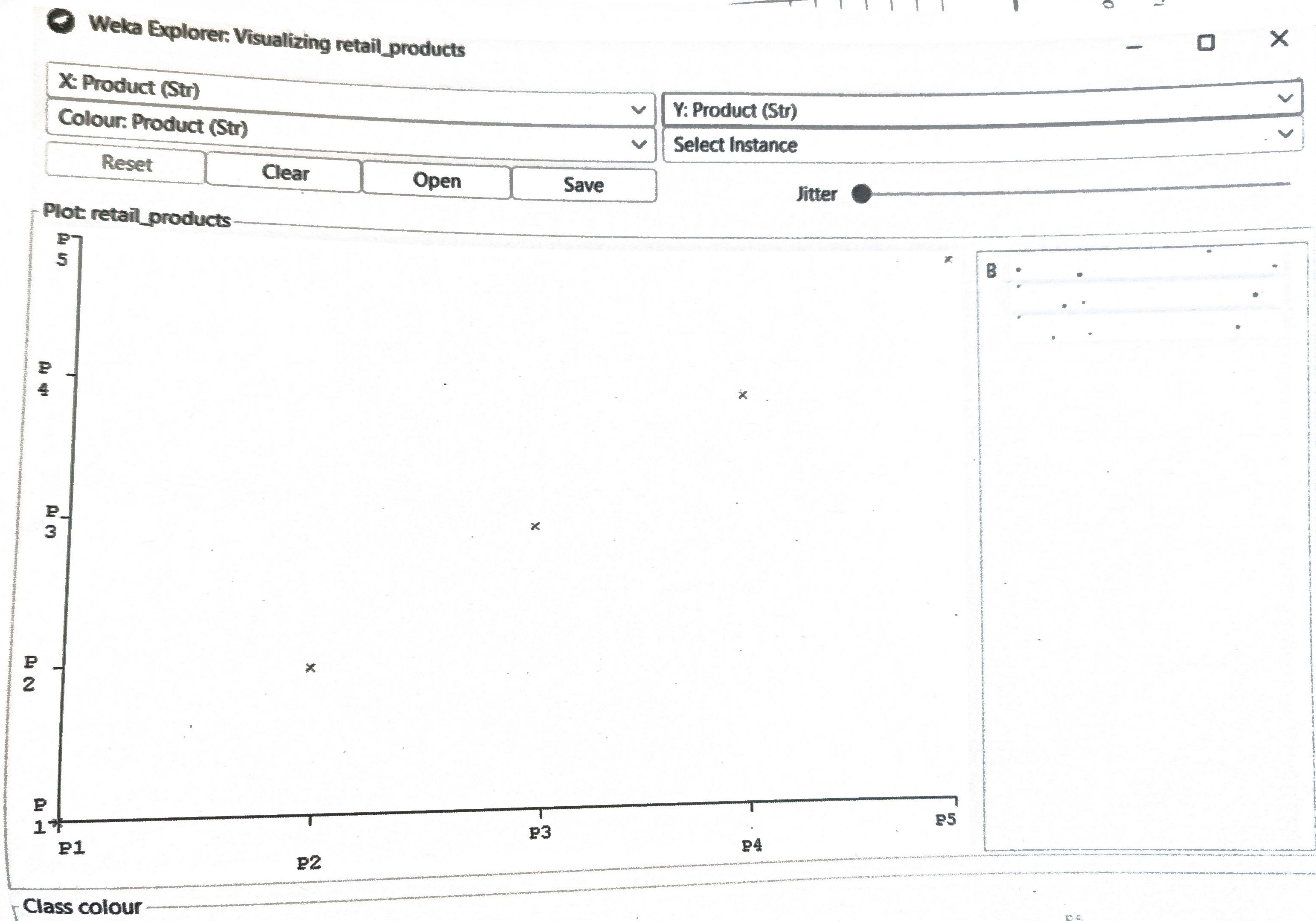
06:58:54 - SimpleKMeans
06:59:45 - SimpleKMeans

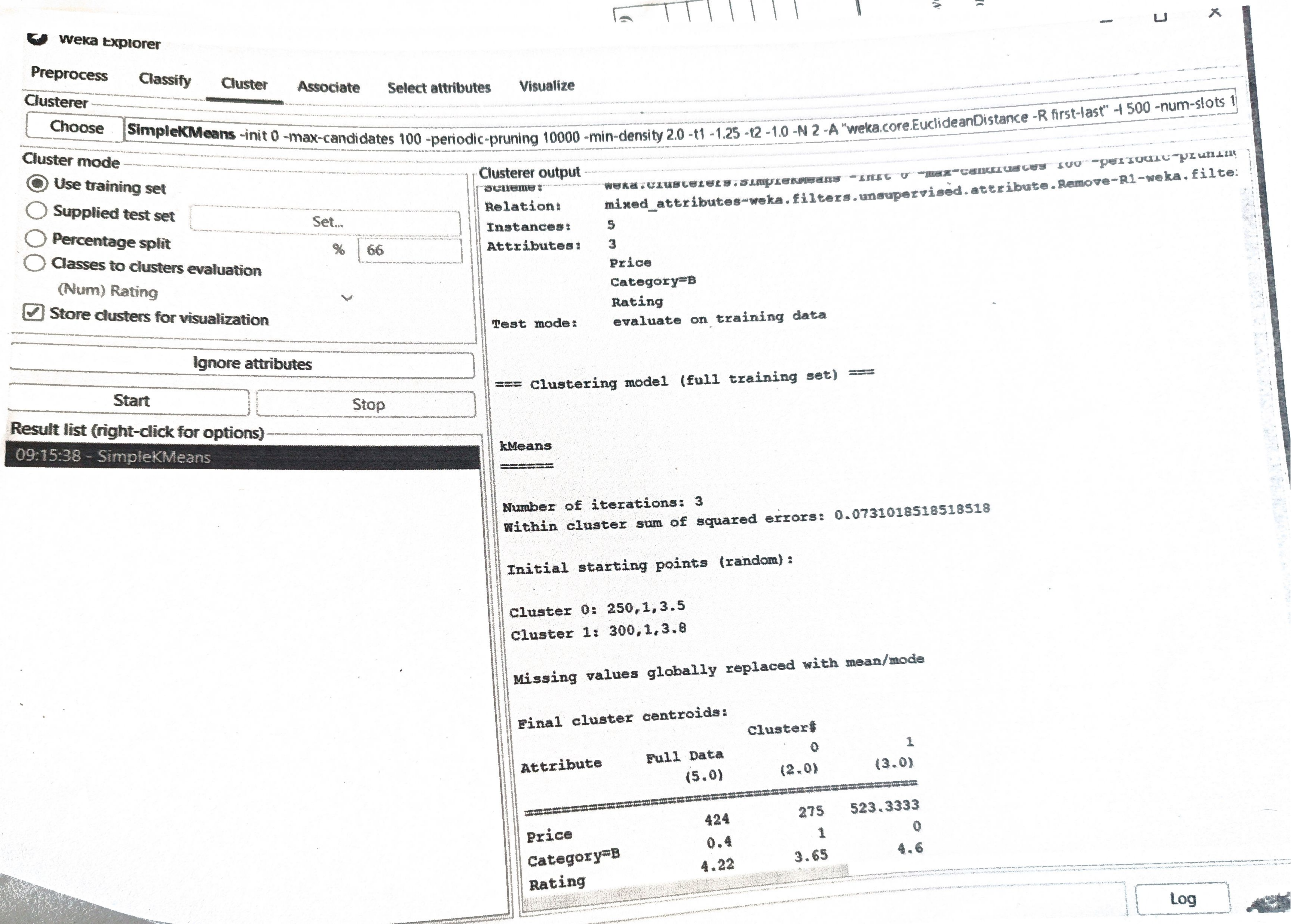
Clusterer output

```
kMeans
======

Number of iterations: 4
Sum of within cluster distances: 1.3153846153845785

Initial starting points (random):

Cluster 0: 0.615385,80.3
Cluster 1: 1,80.25

Missing values globally replaced with mean/mode

Final cluster centroids:
                          Cluster#
Attribute    Full Data          0          1
                 (5.0)      (3.0)      (2.0)
============================================
Latitude        0.6154     0.8462     0.1154
Longitude        80.25      80.27      80.21




Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0       3 ( 60%)
1       2 ( 40%)
```

Log


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
lot Matrix
Product
Sales
Frequency
Frequency
Sales
Product
PlotSize: [100]
PointSize: [1]
Jitter:
Colour: Frequency
Class Colour
Weka Explorer: Visualizing retail_products
X: Product (Str)
Y: Product (Str)
Colour: Sales (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: retail_products
P5
P4
P3
P2
P1
P1
P2
P3
P4
P5
B
Class colour
25000
38500
52000


Weka Explorer: Visualizing retail_products
X: Product (Str)
Y: Product (Str)
Colour: Product (Str)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: retail_products
P5
P4
P3
P2
P1
P1
P2
P3
P4
P5
Class colour


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Rating
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
09:15:38 - SimpleKMeans
Clusterer output
Relation: mixed_attributes-weka.filters.unsupervised.attribute.Remove-R1-weka.filte
Instances: 5
Attributes: 3
Price
Category=B
Rating
Test mode: evaluate on training data
=== Clustering model (full training set) ===
kMeans
======
Number of iterations: 3
Within cluster sum of squared errors: 0.07310185185185185
Initial starting points (random):
Cluster 0: 250,1,3.5
Cluster 1: 300,1,3.8
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Price 424 275 523.3333
Category=B 0.4 1 0
Rating 4.22 3.65 4.6
Log